An empirical instance of set-theoretic estimation: mechanized reconciliation of a multi-voice corpus

A plan — Haiku extracts the frames, Lean verifies the inference

thejeb project note

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Abstract

This note fixes a concrete empirical target for the STE/frames machinery and lays out the pipeline. The task: acquire a corpus of several authors discussing one topic and its internal relations, extract each author's claims into a structured *frame* using a Claude Haiku model, hand the frames to the verified Lean core, and read off the machine-checked invariants (feasible set, concept lattice, obstruction, rectangle-cover complexity, treewidth). The topic chosen is **the cause of the ocean tides**, with a diachronic corpus of seven public-domain natural philosophers spanning antiquity to the early nineteenth century. The architecture keeps the fuzzy natural-language front-end (Haiku) strictly separate from the machine-checked inference core (Lean), with an explicit extraction contract marking the boundary of trust.

1 Why this topic

The demonstration is only compelling if the disagreement in the corpus is *structured*: partly reconcilable, partly irreducibly coupled, with a famous outlier. The cause of the tides is the ideal first instance.

- **Genuine, structured disagreement.** From antiquity the tides were tied to the moon (Pliny the Elder, 77; Seleucus of Seleucia; Bede, 725). Kepler made lunar *attraction* the cause (Kepler, 1609); Descartes replaced attraction with the *pressure* of a celestial vortex (Descartes, 1644); Newton unified sun and moon under *universal gravitation* (Newton, 1687); Laplace gave the dynamical theory (Laplace, 1799). And the famous outlier: Galileo rejected the moon as an "occult" cause and derived the tides from the Earth's combined rotation and revolution (Galilei, 1632, Fourth Day).
- **The coupling is real.** Galileo's frame *requires* the Earth to move and *denies* the moon a role, which contradicts every other voice's affirmation of the lunar correlation. This is exactly a locally-coherent-but-globally-impossible configuration — a nonzero Čech obstruction is expected, with Galileo an isolated concept in the lattice.
- **Public-domain, bounded, legible.** Every source text is out of copyright and obtainable; the topic decomposes into a handful of discrete sub-claims with small domains; and the meaning of each invariant is transparent to a reader.

2 The corpus

Seven voices, assembled into `sources/tides/` with per-item provenance (author, work, date, edition/translation, URL, license). Exact editions and excerpt ranges are pinned in Stage 0.

Voice	Source	Position on the cause
Pliny the Elder	*Naturalis Historia* II (c. 77)	moon+sun correlation
Bede	*De temporum ratione* (725)	lunar timing
Kepler	*Astronomia Nova* (1609)	lunar attraction
Galileo	*Dialogue*, Fourth Day (1632)	Earth's motion; *not* the moon
Descartes	*Principia Philosophiae* (1644)	vortex pressure
Newton	*Principia* III (1687)	universal gravitation (sun+moon)
Laplace	*Mécanique céleste* (1799)	dynamical (gravitational)

3 The topic's relations: the variable schema

The "topic and its relations" becomes a fixed set of variables V with small per-variable domains A_v. Each author's frame is a partial assignment (silence = no constraint on that variable). A candidate schema:

Variable	Domain
`primary_cause`	{moon, sun, moon+sun, earth_motion, vortex, other}
`moon_role`	{attraction, pressure, none}
`sun_role`	{contributes, none}
`mechanism`	{gravitation, aetherial_pressure, kinematic, occult/unspecified}
`requires_earth_motion`	{yes, no}
`spring_neap_phase_link`	{affirmed, denied, silent}
`two_tides_per_day`	{explained, unaddressed}
`quantitative`	{yes, no}

The variables are *coupled* — e.g. `mechanism`=gravitation entails `moon_role`=attraction and `sun_role`=contributes; Galileo's `requires_earth_motion`=yes with `moon_role`=none conflicts with `spring_neap_phase_link`=affirmed. These edges give the relation graph a genuine, nontrivial treewidth — the quantity the tractability theory is about.

4 Architecture: Haiku to the frames, Lean from the frames on

```
[Claude Haiku]  corpus text --> per-author structured frame (strict JSON)
      |         claude-haiku-4-5, output_config.format = json_schema
      |         (fuzzy, unverified: all real-world risk lives here)
      v
[handoff]       V, per-variable domains A_v, per-author constraint S_i
      v
[Lean core]     computable STE instance + adaptive-consistency solver,
      |         proven to refine the verified Set-based spec
      v         outputs, each with a theorem behind it:
 * feasibilitySet = globally consistent readings (or empty)
```

```
* concept lattice = coherent voice-clusters (Ste.HyperFrame)
* cechObstruction / rho = irreducibility of the disagreement
* inducedTreewidth = was reconciliation tractable
* feasible  -> extracted consensus reading (backtrack-free)
* infeasible-> minimal conflicting cores
```

Stage 0 — corpus assembly

Collect the seven excerpts on the tides into `sources/tides/` with a provenance manifest; cite every source in `refs.bib` (done: entries `galileo1632dialogue`, `kepler1609astronomia`, `descartes1644principia`, `newton1687principia`, `laplace1799mecanique`, `pliny77naturalis`, `bede725temporum`).

Stage 1 — frame extraction (Claude Haiku)

For each author-excerpt, one Haiku call with a *strict output schema* returns the frame. Concretely: `claude-haiku-4-5` via the Messages API with `output_config.format` a `json_schema` whose properties are exactly the variables of §4, each an `enum` over its domain plus a `"silent"` option; a `quote` field per assigned variable carrying the sentence that licenses it (auditability). Extraction is cheap ($1/$5 per Mtok; a seven-voice corpus is a handful of calls), and corpus-scale runs use the Batches API at half price. A second normalization pass maps free-text values onto the fixed domain vocabulary. The model does constrained *extraction*, never inference — that is the core's job.

Stage 2 — handoff

The JSON frames become the STE instance: the variable set V, the per-variable domains A_v (the union of asserted values), and one constraint S_i per author (the partial assignment it asserts). This is the exact input shape of the bucket network of `Ste.AdaptiveSolver` (`bucketNetworkList`), whose joint constraint is literally `adaptiveSet`.

Stage 3 — Lean inference (verified core)

A *computable refinement*: a `Finset`/decidable mirror of the Set-based constraint model plus the adaptive-consistency solver, proven to agree with the verified spec, so the aggregation *executes* (`#eval`/compiled) rather than being re-implemented on faith. It reports: the feasible set (globally consistent readings); the concept lattice of coherent voice-clusters (`partialFeasibilitySet_eq_lowerPolar`, `Ste.HyperFrame`); the Čech obstruction and rectangle-cover complexity ρ measuring how irreducibly the authors disagree (`Ste.CechObstruction`, `Ste.RepresentationBounds`); the induced treewidth of the relation graph (`Ste.Treedecomp`); and, when feasible, an extracted consensus reading via backtrack-free extraction (`adaptiveNetwork_solved_by_elimination`) — when infeasible, the minimal conflicting cores.

Stage 4 — writeup

Report the invariants and interpret them, with the honest limits below in plain view.

5 The extraction contract and its limits

The Lean core certifies the *aggregation logic*, not the *reading of the text*. The soundness of the whole is bounded by the natural-language→constraint map of Stage 1: garbage in, *verified* garbage out. Two honest boundaries, to be stated in any writeup:

1. **Extraction fidelity.** Haiku's frame for each author is an interpretation; the `quote` field makes every assigned value auditable against the source sentence, but a wrong reading produces a wrong-but-machine-checked conclusion. The invariants are conditional on the frames.

2. **The computable mirror is real work.** The current modules are specifications (Set-valued, partly `noncomputable`); the decidable refinement and its agreement proof are a bounded but genuine sub-project (Stage 3).

6 The demonstration payoff (predicted)

If the frames come out as the history suggests, the machine-checked invariants should show:

- a **nonzero Čech obstruction** — no single global reading satisfies all seven voices, because Galileo's Earth-motion frame denies the lunar cause the other six require;

- Galileo as an **isolated concept** in the lattice, with the ancient/lunar voices, the attraction voices (Kepler, Newton, Laplace), and the vortex voice (Descartes) forming the coherent clusters;

- a **small** ρ — the disagreement is covered by a few coherent camps, not $2^{|V|}$ arbitrary readings;

- a **low treewidth** on the relation graph — the topic is cheap to reconcile once the outlier is removed, and dropping Galileo's constraint yields a nonempty feasible set (the modern synthesis).

That quantitative verdict — *this corpus's disagreement is coupled through exactly one variable, with obstruction localized to one voice* — machine-certified, is precisely what a pure-embedding or summarization pipeline cannot deliver.

7 Phasing

P0. Stage 0 corpus + provenance manifest; bib done.

P1. Fix the variable schema; write and validate the Haiku extraction prompt + JSON schema on the seven excerpts.

P2. Build the computable Lean core (`Finset` refinement + agreement proof); CI-green.

P3. Wire Stage 2 handoff; run Stage 3; produce the invariant table.

P4. Writeup with the honest-limits section; build the PDF.

References

Bede. *De temporum ratione.* 725. Ties tidal timing to the moon; the medieval "establishment of port".

René Descartes. *Principia Philosophiae.* 1644. Part IV: tides as the pressure of the celestial vortex.

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Pierre-Simon Laplace. *Traité de mécanique céleste*. 1799. Dynamical theory of the tides (1799–1825).

Isaac Newton. *Philosophiae Naturalis Principia Mathematica*. 1687. Book III: tides from the gravitation of sun and moon; trans. Andrew Motte, 1729.

Pliny the Elder. *Naturalis Historia*. 77. Book II records the correlation of tides with the moon and sun.